1.(a)	States/uses either $16k=-4$ or $\frac{16\times15}{2}k^2=p$	M1
	(i) $k=-\frac{1}{4}$	A1
	(ii) $p=\frac{15}{2}$	A1
		(3)
(b)	$g(x)=\left(2+\frac{16}{x}\right)(1+kx)^{16}$	
	Attempts either $2"p"$ or $16\times\frac{16\times15\times14}{3!}\times k^3$	M1
	Attempts sum of $2"p"$ and $16\times\frac{16\times15\times14}{3!}\times k^3$	dM1
	Term in $x^2=(15-140)x^2=-125x^2$	A1
		(3)
		(6 marks)

2.(a)	Attempts to substitute $u_2 = 6k+3$ in $u_3 \left(= ku_2 + 3\right)$ $u_3 = k\left(6k+3\right)+3$	M1 A1 **(2)**
(b)	Uses $\sum_{n=1}^{3} u_n = 117 \Rightarrow 6+6k+3+k(6k+3)+3=117$ $6k^2+9k-105=0 \Rightarrow k=...$ $k=\frac{7}{2}$	M1 dM1 A1 **(3)**
		(5 marks)

3. (a)	States or uses $h = 3$	B1
	Attempts $\frac{"3"}{2}\left\{\log_{10} 2 + \log_{10} 14 + 2\times(.....)\right\}$	M1
	$= \frac{3}{2}\left\{\log_{10} 2 + \log_{10} 14 + 2\times\left(\log_{10} 5 + \log_{10} 8 + \log_{10} 11\right)\right\} = 10.10$ (2dp)*	A1*
		(3)
(b)	Increase the number of strips	B1
		(1)
(c) (i)	$\int_2^{14} \log_{10}\sqrt{x}\ \mathrm{d}x = \frac{1}{2}\times 10.10 = 5.05$	B1
(ii)	$\log_{10} 100x^3 = 2 + 3\log_{10} x$	B1
	$\int_2^{14} \log_{10} 100x^3\ \mathrm{d}x = \left[2x\right]_2^{14} + 3\times 10.10 = 54.30$	M1 A1
		(4)
		(8 marks)

4.(a)	States -21	B1 **(1)**
(b)	Attempts $f(3)=\left(3^2-2\right)\left(2\times3-3\right)-21$	M1
	Achieves $f(3)=0\Rightarrow\left(x-3\right)$ is a factor of $f(x)$ *	A1* **(2)**
(c) (i)	$f(x)=2x^3-3x^2-4x-15=\left(x-3\right)\left(2x^2....\pm5\right)$	B1 M1
	$=\left(x-3\right)\left(2x^2+3x+5\right)$	A1
(ii)	Attempts b^2-4ac for their $2x^2+3x+5$	M1
	Achieves $b^2-4ac<0$ and states that only root is $x=3$ o.e. *	A1* **(5)** **(8 marks)**

5.(a)	Attempts $480+13\times-15$ = 285 tonnes	M1 A1 **(2)**
(b)	Sets $\frac{N}{2}\{2\times480+(N-1)\times-15\}=7770$ $\frac{N}{2}\{960-15N+15\}=7770\Rightarrow N^2-65N+1036=0$ *	M1A1 A1* **(3)**
(c)	States 28 only	B1 **(1)** **(6 marks)**

6 (i)	$x^2 + y^2 + 10x - 12y = k$	
(a)	Attempts $(x \pm 5)^2 + (y \pm 6)^2 ..... = 0$	M1
	Centre $(-5, 6)$	A1
		(2)
(b)	Sets $k + (\pm 5)^2 + (\pm 6)^2 > 0$	M1
	$k > -61$	A1
		(2)
(ii)	Centre $\left(\frac{-2+8}{2}, \frac{10+(-14)}{2}\right) = ...$; radius $\frac{1}{2}\sqrt{(-2-8)^2 + (10-(-14))^2} = ...$	M1
	Centre is $(3, -2)$; radius is 13	A1
	$\Rightarrow C_2 : (x-3)^2 + (y+2)^2 = 13^2 \Rightarrow (p-3)^2 + 4 = 169 \Rightarrow p = ...$ Or $PX = r \Rightarrow (p-3)^2 + (0-(-2))^2 = 169 \Rightarrow p = ...$	M1
	$p = 3 + \sqrt{165}$ only	A1
		(4)
		(8 marks)

Alt 6 (ii)	Uses either $PQ^2 + PR^2 = RQ^2$ or grad $PQ \times$ grad $PR = -1$	M1
	Correct equation. E.g. $(p+2)^2 + 10^2 + (p-8)^2 + 14^2 = (-2-8)^2 + (10+14)^2$ Alternatively $\frac{-10}{p+2} \times \frac{14}{p-8} = -1$	A1
	Correct method to set up and solve 3TQ. FYI $p^2 - 6p - 156 = 0$	dM1
	$p = 3 + \sqrt{165}$ only	A1
		(4)

7. (i)	States or implies that $4\times6^{n-1}>10^{100}$	B1
	Takes logs correctly to produce an equation without powers $4\times6^{n-1}>10^{100}\Rightarrow\log4+(n-1)\log6>100\log10$	M1
	$n=129$	A1
		(3)
(ii) (a)	States $ar=-6$ and $\dfrac{a}{1-r}=25$	B1
	Combines to form an equation in $r\Rightarrow\dfrac{-6}{r(1-r)}=25$	M1
	$\Rightarrow-6=25r(1-r)\Rightarrow25r^2-25r-6=0$ *	A1*
		(3)
(b)	$r=\dfrac{6}{5},-\dfrac{1}{5}$	B1
		(1)
(c)	$r=-\dfrac{1}{5}$ as $\lvert r\rvert<1$ (for S_∞ to exist)	B1
		(1)
(d)	Attempts $S_4=\dfrac{a\left(1-r^n\right)}{1-r}$ with $n=4, r=\text{their}(c)$ and $a=\dfrac{-6}{\text{their}(c)}$	M1
	$S_4=\dfrac{30\left(1-\left(-\dfrac{1}{5}\right)^4\right)}{1-\left(-\dfrac{1}{5}\right)}=24.96$ o.e.	A1
		(2)
		(10 marks)

8 (a)	$y = \frac{4}{3}x^3 - 11x^2 + kx \Rightarrow \frac{dy}{dx} = 4x^2 - 22x + k$	M1
	uses $x = 2, \frac{dy}{dx} = 0 \Rightarrow 0 = 16 - 44 + k \Rightarrow k = 28$ *	dM1 A1*
		(3)
(b)	$\frac{dy}{dx} = 4x^2 - 22x + 28 = 0 \Rightarrow (2x-4)(2x-7) = 0 \Rightarrow x = ...$	M1
	$x < 2, x > \frac{7}{2}$	A1
		(2)
(c)	$\int \left(\frac{4}{3}x^3 - 11x^2 + 28x\right) dx \Rightarrow \frac{1}{3}x^4 - \frac{11}{3}x^3 + 14x^2$	M1 A1
	Correct y coordinate of $M = \frac{68}{3}$	B1
	Complete method to find $R = 2 \times "\frac{68}{3}" - \int_0^2 \left(\frac{4}{3}x^3 - 11x^2 + 28x\right) dx$	M1
	$= 2 \times "\frac{68}{3}" - \left(\frac{1}{3} \times 2^4 - \frac{11}{3} \times 2^3 + 14 \times 2^2\right)$	
	$= \frac{40}{3}$	A1
		(5)
		(10 marks)

9.(a)	(If x and y are positive) $\left(\sqrt{x}-\sqrt{y}\right)^2 \ldots 0 \Rightarrow x-\ldots\sqrt{xy}+y \ldots 0$ $\Rightarrow x-2\sqrt{xy}+y \ldots 0$ $\Rightarrow \frac{x+y}{2} \ldots \sqrt{xy}$	M1 A1 A1* **(3)**
(b)	States for example when $x=-8, y=-2$, $\frac{x+y}{2}=-5$, $\sqrt{xy}=4$ so $\frac{x+y}{2} \leqslant \sqrt{xy}$	B1 **(1)** **(4 marks)**

9.(a) Alt 1	$\frac{x+y}{2} \ldots \sqrt{xy} \Rightarrow \frac{(x+y)^2}{4} \ldots xy \Rightarrow \frac{x^2 + \ldots xy + y^2}{4} \ldots xy$ $\Rightarrow x^2 - 2xy + y^2 \ldots 0 \Rightarrow (x-y)^2 \ldots 0$ States both of the following o.e • $(x-y)^2 \ldots 0$ as it is a square number • so $\frac{x+y}{2} \ldots \sqrt{xy}$ is true	M1 A1 A1* **(3)**

9.(a) Alt 2	States $(x-y)^2 \ldots 0 \Rightarrow x^2 - 2xy + y^2 \ldots 0$	M1
	Rearranges $\Rightarrow x^2 + 2xy + y^2 \ldots 4xy \Rightarrow (x+y)^2 \ldots 4xy$	A1
	States that as x, y positive, so $x+y > 0$ (and $xy > 0$) $\Rightarrow (x+y) \ldots 4\sqrt{4xy} \Rightarrow \frac{x+y}{2} \ldots 4\sqrt{xy}$	A1*
		(3)

10 (i)	$\tan^2\left(2x+\frac{\pi}{4}\right)=3 \Rightarrow 2x+\frac{\pi}{4}=\pm\frac{\pi}{3},\pm\frac{2\pi}{3}$	M1
	$\Rightarrow x=\left(...-\frac{\pi}{4}\right)\div 2$	dM1
	$\Rightarrow x=\frac{\pi}{24},-\frac{11\pi}{24};\frac{5\pi}{24},-\frac{7\pi}{24}$	A1; A1, A1
		(5)
(ii)	$\left(2\sin\theta-\cos\theta\right)^2=1 \Rightarrow 4\sin^2\theta-4\sin\theta\cos\theta+\cos^2\theta=1$	
	Attempts to use $\sin^2\theta+\cos^2\theta=1 \Rightarrow 4\sin^2\theta-4\sin\theta\cos\theta=\sin^2\theta$	M1
	$\Rightarrow \sin\theta\left(3\sin\theta-4\cos\theta\right)=0$	dM1
	$\tan\theta=\frac{4}{3}$	A1
	$\theta=53.1°,233.1°,180°$	A1, <u>B1</u>
		(5)
		(10 marks)

10(ii) Alt	$(2\sin\theta-\cos\theta)^2=1\Rightarrow 2\sin\theta-\cos\theta=\pm1$ $\Rightarrow 2\sin\theta=\cos\theta\pm1$ $\Rightarrow 4\sin^2\theta=\cos^2\theta\pm2\cos\theta+1$ then attempts to use $\sin^2\theta+\cos^2\theta=1\Rightarrow 4-4\cos^2\theta=\cos^2\pm2\cos\theta+1$ $\Rightarrow 5\cos^2\theta\pm2\cos\theta-3=0\Rightarrow\cos\theta=...$ $\cos\theta=\frac{3}{5},-1$ or $\cos\theta=-\frac{3}{5},1$ $\theta=53.1°,233.1°,180°$	M1 dM1 A1 A1, <u>B1</u> **(5)** **(10 marks)**